

Are tropical cyclones still behaving the way risk models assume?

For a reinsurer deciding whether historical cyclone baselines still reflect today's risk

Data and scope

Used storm-track data to measure how cyclone intensity and escalation have changed, basin by basin, since 1988.

TropiCycloneNet – tropical cyclone track data

What it contains	How I used it
Storm location	Track storms by basin and region
Wind speed	Measure intensity and high-wind events
Date and time	Build annual trends and time-series metrics

Analysis scope

Period: 1988–2023 (satellite era)

Basins: Western Pacific, Eastern Pacific, North Atlantic, South Indian, South Pacific

Excluded: North Indian basin due to coverage and consistency concerns

How I tested whether the baseline has shifted

What I measured

I converted 6-hourly storm-track records into basin-year indicators covering:

Severity:	How often storms reach high-intensity thresholds
Rapid Intensification:	How often storms rapidly intensify within 24 hours
Movement:	Average storm movement speed

How I tested for change

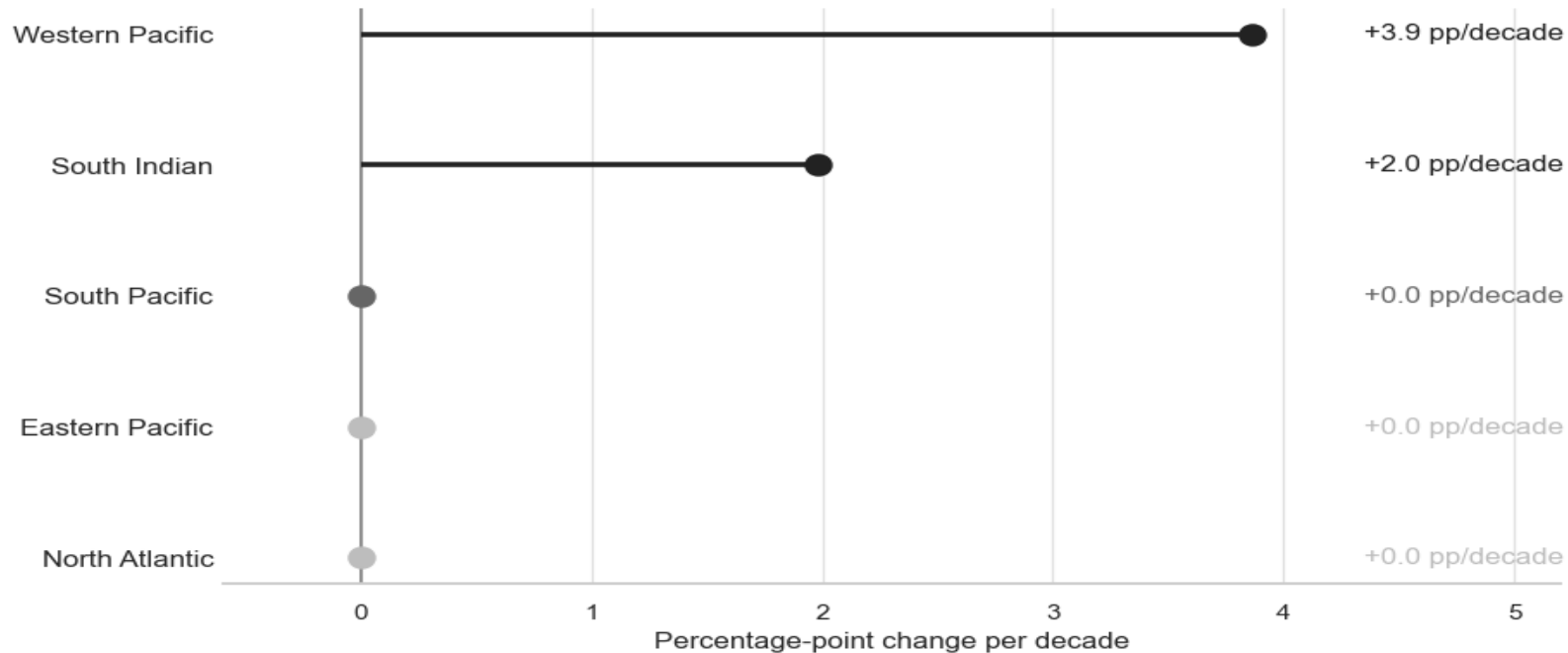
Annual trends across 1988–2023, tested per basin using rank-based trend tests to separate genuine shifts from year-to-year variability.

What this approach cannot tell us

This is a diagnostic of past change, not a forecast. It identifies what has shifted in the historical record, not what will happen next.

Severity

The high-intensity tail has shifted upward in Western Pacific and South Indian, but not in other basins.
Annual share of storms with peak sustained wind ≥ 58.1 m/s, by basin, 1988–2023



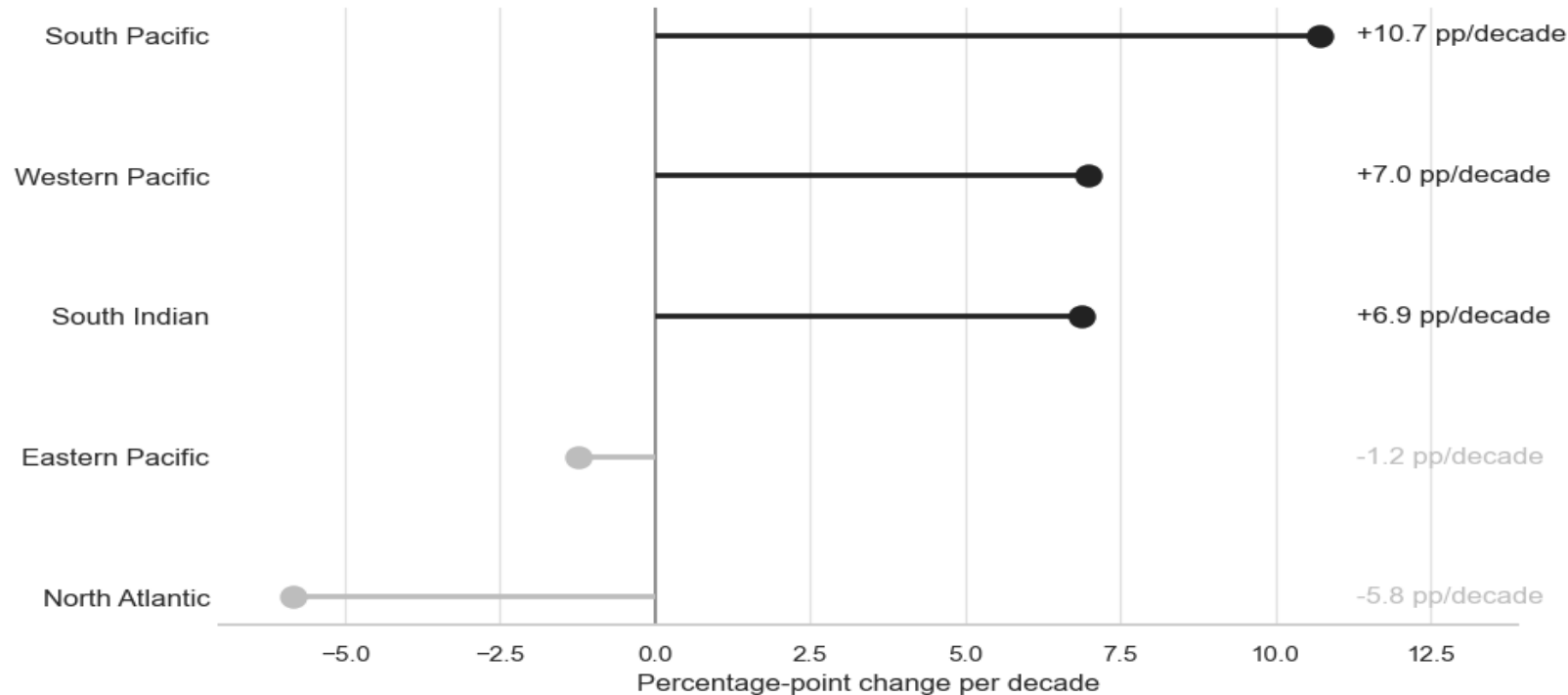
Annual trend tests show statistically meaningful upward shifts in the Western Pacific and South Indian basins. South Pacific is treated as a sensitivity case, while Eastern Pacific and North Atlantic show no meaningful upward trend.

Action: Before renewal pricing, stress-test South Indian and Western Pacific portfolios against a higher high-intensity event frequency. Treat South Pacific as a case for monitoring, not as a primary pricing trigger.

Rapid Intensification

The share of storms that rapidly intensify has shifted upward in South Pacific, Western Pacific and South Indian, but not in other basins.

Annual share of storms experiencing rapid intensification, by basin, 1988–2023



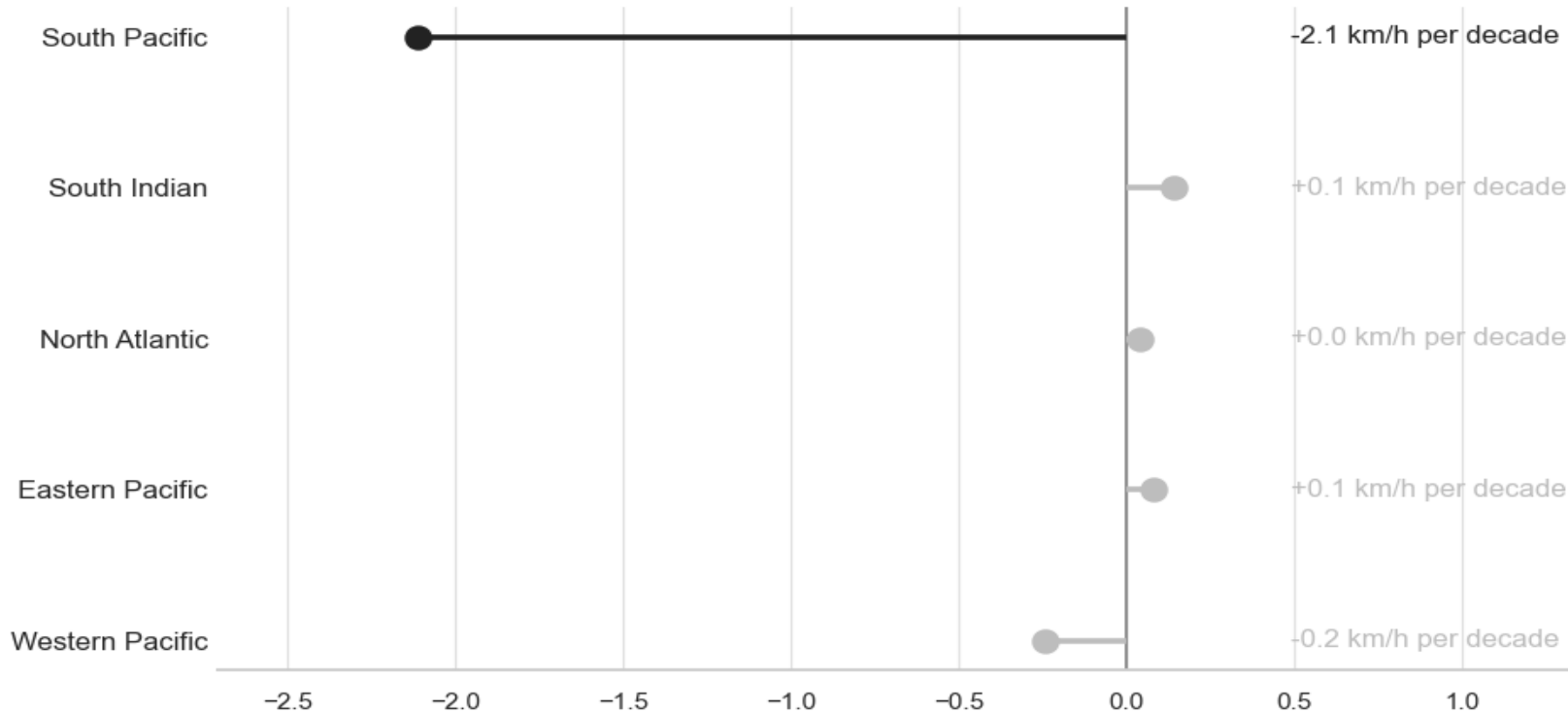
Annual trend tests show statistically meaningful upward shifts in rapid intensification share in South Pacific, Western Pacific and South Indian. Eastern Pacific shows no meaningful trend. North Atlantic is borderline downward, not a pricing trend.

Action: Before renewal pricing, stress-test South Pacific, Western Pacific and South Indian portfolios for shorter warning windows and higher loss volatility from rapidly intensifying storms.

Movement

Storm movement has slowed meaningfully only in South Pacific.

Annual mean storm translation speed, by basin, 1988–2023



Annual trend tests show a statistically meaningful decline in South Pacific translation speed. Other basins show no meaningful movement trend.

Action: For South Pacific portfolios, stress-test slower-moving storm scenarios: longer local exposure, higher rainfall accumulation, and longer business interruption.

Synthesis and recommendations

Three risk channels reach the same conclusion: update cyclone baselines basin by basin.

South Pacific appears across all three channels: a severity sensitivity case, the strongest rapid-intensification signal, and the only clear movement-speed slowdown.

Channel	Recommended action
Severity	Stress-test Western Pacific and South Indian for higher frequency of high-intensity events.
Rapid Intensification	Stress-test South Pacific, Western Pacific and South Indian for shorter warning windows and higher loss volatility.
Movement	Stress-test South Pacific for slower-moving storms scenarios: longer local exposure, higher potential rainfall accumulation and longer business interruption.

Bottom line: A single global cyclone baseline risks missing regional shifts in current hazard behaviour. Move to basin-specific monitoring before renewal pricing, capital, or attachment-point decisions are finalised.

Limitations and next steps

Limitations

- This is a diagnostic of past change, not a forecast.
- Results are based on a single dataset (TropiCycloneNet) and have not been cross-validated against alternative cyclone records such as IBTrACS.
- Some basin-metric combinations are sparse, especially South Pacific severity and movement, which makes those findings more sensitive to individual years.

Next steps

- Cross-validate the same trends against IBTrACS.
- Apply count-based models to sparse-data basins where many years have zero high-intensity storms.
- Test alternative baseline windows, including a post-2010 view, before any pricing or capital assumptions.